

Linux Fundamentals

18 Practical Exercises - File and Directory Management

Completed: April 2026

Environment: Rocky Linux 9.7 | CentOS-based | RHEL-compatible

Executive Summary

Successfully completed 18 comprehensive exercises focused on Linux file system fundamentals and directory management. This hands-on training aligns with industry-standard certifications such as LFCS (Linux Foundation Certified System Administrator) and RHCSA (Red Hat Certified System Administrator). All exercises were completed independently on a Rocky Linux 9.7 virtual machine, demonstrating practical troubleshooting and system administration skills essential for professional NOC operations and system administration roles.

Topics Covered

Category	Topics	Exercises
File System Basics	Root directory (/), Filesystem tree, Working directories (pwd, cd)	1-4
Path Navigation	Absolute vs Relative paths, Directory specifications	5
Directory Management	Creating directories (mkdir), Parent directory creation (-p flag)	6, 17
File Operations	Creating files (touch), Copying files and directories (cp, cp -r)	7-8
File Preservation	Preserving file attributes (-p flag), Recursive operations	9-10
File Deletion	Removing files (rm), Removing directories (rm -r)	11, 13
File Movement	Moving files and contents (mv), Renaming operations	12, 16
Link Management	Symbolic links (ln -s), Hard links (ln)	14-15
Advanced Listing	Time display options (--full-time), Modification timestamps	18

Exercise Details & Key Learnings

- 1-4 | File System Fundamentals:** Mastered Linux directory hierarchy starting from root (/) and learned navigation commands (pwd, cd). Understanding the filesystem tree structure is foundational for all system administration tasks.
- 5 | Path Specification:** Practiced absolute path notation (/home/bob/Documents) for precise file location identification, essential for scripting and automation.
- 6-7 | Directory & File Creation:** Created directories and files using mkdir and touch commands, understanding file creation from command line.
- 8-10 | Recursive Operations:** Mastered cp -r for copying entire directory structures while preserving attributes with -p flag. Critical skill for backup and migration operations.
- 11-13 | File Deletion:** Safely deleted files with rm and directories with rm -r, understanding the risks and importance of careful file management in production environments.
- 14-15 | Link Management:** Configured both symbolic links (ln -s) and hard links (ln), understanding differences and use cases for each type.
- 16 | File Renaming:** Used mv command for renaming operations, demonstrating file movement capabilities.
- 17-18 | Advanced Options:** Implemented mkdir -p for parent directory creation and explored ls --full-time for exact timestamp display, demonstrating exploration of man pages and command-line options.

Competencies Acquired

✓ Command Line Proficiency	✓ File System Navigation	✓ Automation Ready
✓ Man Page Research	✓ Recursive Operations	✓ Path Specification
✓ Link Creation & Management	✓ Attribute Preservation	✓ Troubleshooting Skills

Path to LFCS & RHCSA Certification

These exercises form the **foundational knowledge base** required for both LFCS and RHCSA certifications. This hands-on experience with file system operations, command-line navigation, and directory management demonstrates readiness for advanced Linux administration topics including user/group management, permissions, networking, system services, and security. Continued practice in a production-like environment (Rocky Linux 9.7) ensures practical competency aligned with industry standards.